Supporting Information

The Universal Calibration for STRUCTURE- and SOLVENT-Independent Molar Mass Determinations of Polymers using Diffusion-Ordered Spectroscopy

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Theoretical Basis

intrinsic viscosity according to Flory:[1]

$$[\eta] = \frac{2.5\ N_A\, V_h}{M} \tag{1}$$

Mark-Houwink relation:[2]

$$[\eta] = K\ M^{a} \tag{2}$$

From equation (1) and (2) follows equation (3):

$$V_h = \frac{1}{2.5 * N_A} K\ M^{a+1} \tag{3}$$

$$V_h = \frac{4}{3}\pi\ r_h^3 \tag{4}$$

From equation (3) and (4) follows equation (5):

$$K\ M^{a+1} = \frac{10}{3}\pi\ N_A\ r_h^3 \tag{5}$$

Stokes-Einstein equation:[3]

$$r_h = \frac{k_B T}{6\ \pi\ \eta\ D} \tag{6}$$

including (6) into (5) yields equation (7):

$$K\ M^{a+1} = \frac{10}{3}\ \pi\ N_A \left(\frac{k_B\ T}{6\ \pi\ \eta\ D}\right)^3 \tag{7}$$

For the determination of K_2 and a_2 the following proportionalities for a given T are used:

$$K_1\ M_1^{a_1+1} \sim \frac{1}{\eta_1^3 D_1^3} \tag{8a}$$

$$K_2\ M_2^{a_2+1} \sim \frac{1}{\eta_2^3 D_2^3} \tag{8b}$$

The ratio of (8a) and (8b) results in (9):

$$\frac{K_1 M_1^{a_1+1}}{K_2 M_2^{a_2+1}} = \left(\frac{\eta_2 D_2}{\eta_1 D_1}\right)^3 \tag{9}$$

from Rouse-Zimm model derived:[4,5]

$$D = c\ M^{-b} \tag{10}$$

Resolving (10) to the molar mass yields (11):

$$M = \left(\frac{D}{c}\right)^{-1/b} \tag{11}$$

Including (11) in (9) results in (12):

$$\frac{K_1(\frac{D_1}{c_1})^{-(a_1+1)/b_1}}{K_2(\frac{D_2}{c_2})^{-(a_2+1)/b_2}} = \frac{\eta_2^3 D_2^3}{\eta_1^3 D_1^3} \quad (12)$$

Resolving (12) to D_2 is possible by using logarithmic rules:

$$\log(D_2) = A + B\log(D_1) \quad (13)$$

with

$$A = \frac{b_2}{3b_2 - a_2 - 1}\{\log\left(\frac{K_1\eta_1^3}{K_2\eta_2^3}\right) + \frac{a_1+1}{b_1}\log(c_1) - \frac{a_2+1}{b_2}\log(c_2)\}$$

$$B = \frac{b_2(3b_1 - a_1 - 1)}{b_1(3b_2 - a_2 - 1)}$$

D_1 corresponds to the diffusion coefficients of the reference polymer in dependence of M_w. D_2 corresponds to the diffusion coefficients of the second polymer in dependence of M_w where the parameters K_2 and a_2 will be determined.

Tables

Table S1. Molar mass distributions of PS-, PEO and PMMA homopolymer samples used for the calibration.

Polymer	Sample	M_p	M_w	M_n	PDI
PS	ps1.2k	1306	1220	1120	1.09
	ps3.2k	3500	3510	3350	1.05
	ps9k	8680	8670	8420	1.03
	ps18k	19700	19100	18200	1.05
	ps33k	34800	34000	32700	1.04
	ps62k	66000	62500	59300	1.05
PEO	peg1k	1030	1020	978	1.04
	peg3k	3450	3450	3340	1.03
	peg6k	6530	6200	5860	1.06
	peg12k	12600	11400	10300	1.11
	peg26k	26100	25800	22100	1.17
	peg42k	42700	40100	30700	1.31
PMMA	mm2.1k	2160	2130	1920	1.11
	mm6.5k	5980	5930	5590	1.06
	mm12.5k	12900	12800	12300	1.04
	mm21k	22800	21200	18700	1.13
	mm40k	41400	40300	38100	1.06
	mm85k	88500	86700	83700	1.04

Table S2. Pulsed field gradient conditions for all samples and solvents: Gradient pulse length given as little delta (LD) and diffusion delay as big delta (BD) for the convection compensated pulse sequence dstebpgp3s.

Sample	Gradients	$CDCl_3$	C_6D_6	CD_2Cl_2	Toluene	THF	Acetone
ps1.2k	LD/ms	2.0	2.0	1.6	2.0	2.0	1.6
	BD/ms	76.73	76.73	77.27	76.73	76.73	57.27
ps3.2k	LD/ms	2.0	2.0	1.6	2.0	2.0	1.6
	BD/ms	76.73	76.73	77.27	76.73	76.73	57.27
ps9k	LD/ms	2.6	2.6	2.4	2.6	2.6	2.0
	BD/ms	95.93	95.93	76.2	95.93	95.93	76.73
ps18k	LD/ms	3.2	3.2	3.0	3.2	3.2	2.6
	BD/ms	95.13	95.13	95.4	95.13	95.13	75.93
ps33k	LD/ms	3.8	3.8	3.6	3.8	3.8	3.0
	BD/ms	94.33	94.33	94.6	94.33	94.33	75.4
ps62k	LD/ms	4.2	4.2	4.0	4.2	4.2	-
	BD/ms	113.8	113.8	94.07	113.8	113.8	-
Sample	**Gradients**	**MeOD**	**D_2O**	**CD_3CN**	**Acetone**	**THF**	
peg1k	LD/ms	2.0	2.6	1.6	1.6	2.0	
	BD/ms	76.73	75.93	77.27	57.27	76.73	
peg3k	LD/ms	2.0	2.6	1.6	1.6	2.0	
	BD/ms	76.73	75.93	77.27	57.27	76.73	
peg6k	LD/ms	2.6	3.2	2.4	2.0	2.4	
	BD/ms	95.93	95.13	76.2	76.73	96.2	
peg12k	LD/ms	3.2	3.8	3.0	2.6	3.0	
	BD/ms	95.13	94.33	95.4	75.93	95.4	
peg26k	LD/ms	3.8	4.6	3.6	3.0	3.6	
	BD/ms	94.33	93.27	94.6	75.4	94.6	
peg42k	LD/ms	4.2	5.2	4.0	-	4.0	
	BD/ms	113.8	112.47	94.07	-	114.07	
Sample	**Gradients**	**$CDCl_3$**	**C_6D_6**	**CD_2Cl_2**	**DMSO**	**THF**	
mm2.1k	LD/ms	2.0	2.0	1.6	3.6	2.0	
	BD/ms	76.73	76.73	77.27	74.6	76.73	
mm6.5k	LD/ms	2.0	2.0	2.0	3.8	2.2	
	BD/ms	76.73	76.73	76.73	94.33	76.47	
mm12.5k	LD/ms	2.6	2.6	2.4	4.0	2.6	
	BD/ms	95.93	95.93	96.2	94.07	95.93	
mm21k	LD/ms	3.2	3.2	3.0	4.6	3.2	
	BD/ms	95.13	95.13	95.4	113.27	95.13	
mm40k	LD/ms	3.8	3.8	3.6	5.2	3.8	
	BD/ms	94.33	94.33	94.6	122.47	94.33	
mm85k	LD/ms	4.2	4.2	4.0	5.6	4.2	
	BD/ms	113.8	113.8	94.07	141.93	113.8	

Table S3. Comparison of Mark-Houwink parameters **K** and **a** determined by DOSY and provided by available literature data (rounded to the same number of digits).

Polymer	Solvent	K (DOSY)	a (DOSY)	K (literature)	a (literature)	literature
PS	Benzene	0.01077	0.727	0.0113	0.73	[6]
	Chloroform	0.00732	0.746	0.00716	0.76	[6]
	Toluene	0.00980	0.735	0.00977	0.73	[6]
	THF	0.01410	0.700	0.0141	0.7	[6]
PEO	Water	0.02554	0.678	0.04683	0.659	[7]
	Acetone	0.06862	0.565	0.156	0.5	[2]

	Benzene	0.00848	0.733	0.00745	0.76	[8]
PMMA	Chloroform	0.00517	0.771	0.00581	0.79	[8]
	THF	0.01061	0.695	0.0104	0.697	[8]

References

(1) Flory, P. J. *Principles of Polymer Chemistry*; Cornell University Press: Ithaka, N. Y., 1953.

(2) Rudin, A.; Johnston, H. K. Diffusion Coefficients and Intrinsic Viscosities of Polymers. *J Polym Sci B* **1971**, *9* (1), 55–60. https://doi.org/10.1002/pol.1971.110090109.

(3) Einstein, A. Über Die von Der Molekularkinetischen Theorie Der Wärme Geforderte Bewegung von in Ruhenden Flüssigkeiten Suspendierten Teilchen. *Ann Phys* **1905**, *322* (8), 549–560. https://doi.org/10.1002/andp.19053220806.

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(7) Rauschkolb, J. C.; Ribeiro, B. C.; Feiden, T.; Fischer, B.; Weschenfelder, T. A.; Cansian, R. L.; Junges, A. Parameter Estimation of Mark-Houwink Equation of Polyethylene Glycol (PEG) Using Molecular Mass and Intrinsic Viscosity in Water. *Biointerface Res Appl Chem* **2021**, *12* (2), 1778–1790. https://doi.org/10.33263/BRIAC122.17781790.

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Figures

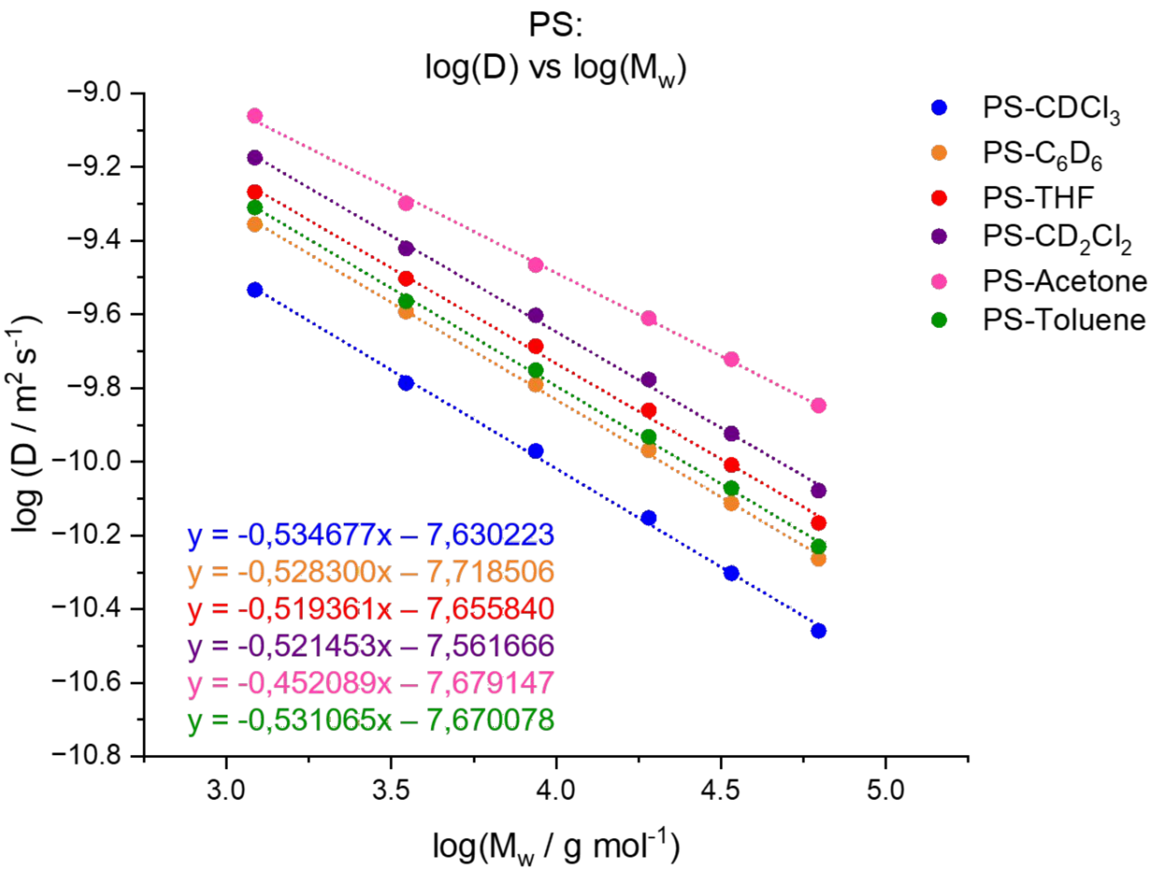

Figure S1. Diffusion coefficients in dependence of the molar mass for polystyrene in different solvents.

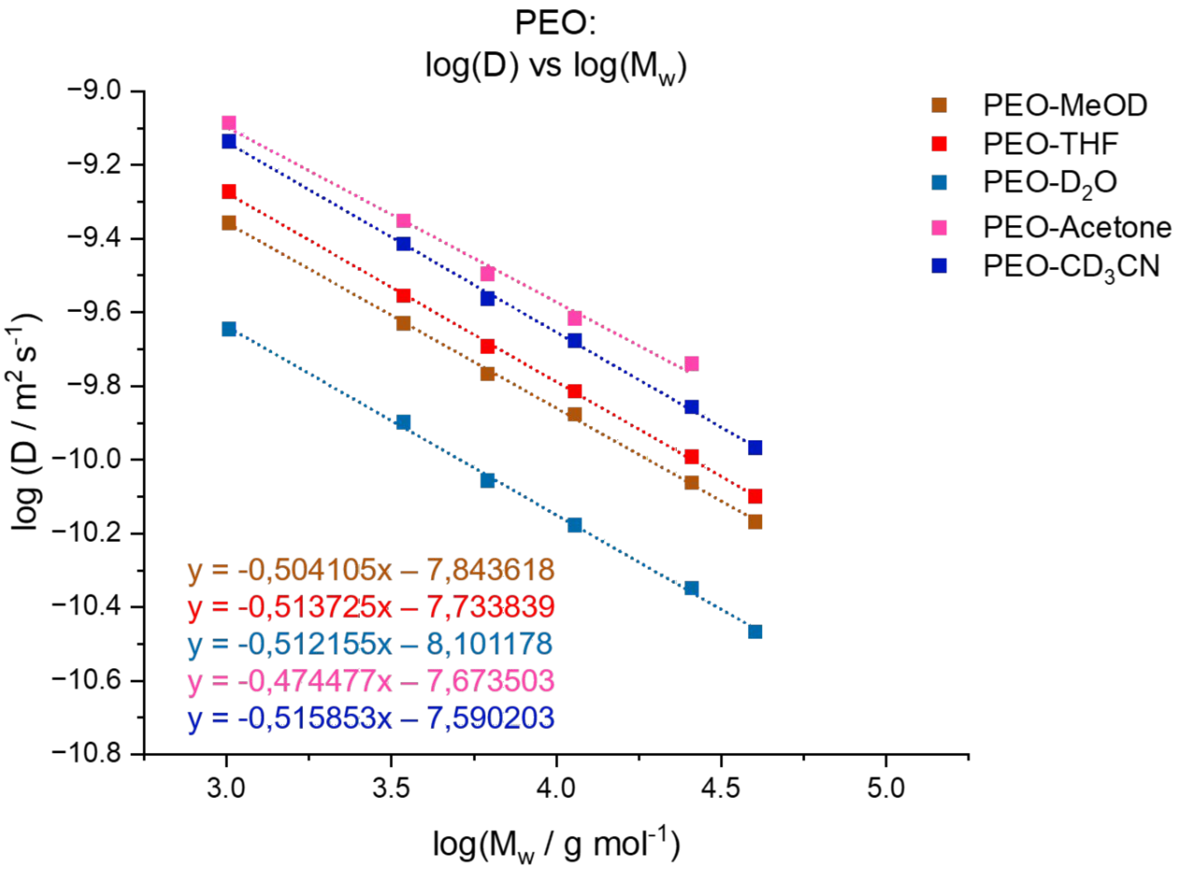

Figure S2. Diffusion coefficients in dependence of the molar mass for polyethylene oxide in different solvents.

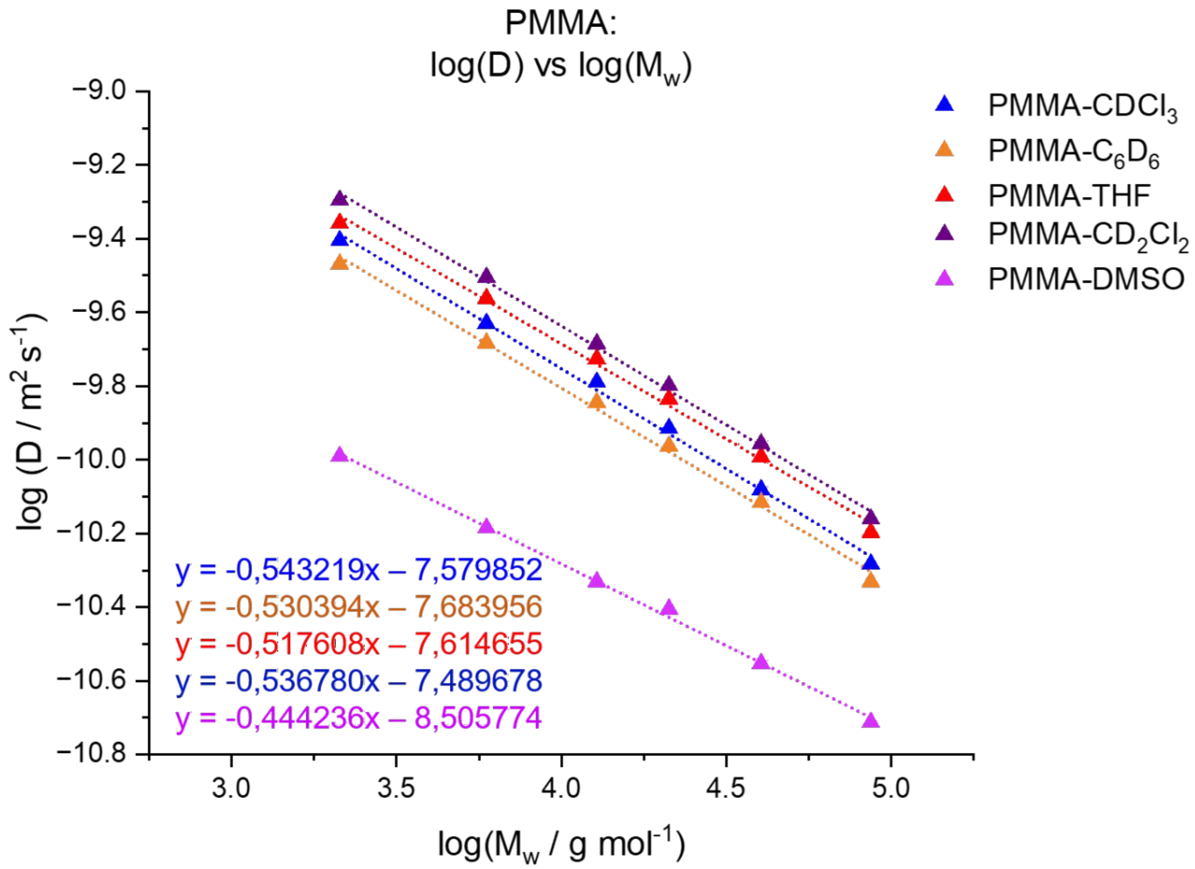

Figure S3. Diffusion coefficients in dependence of the molar mass for poly(methyl methacrylate) in different solvents.

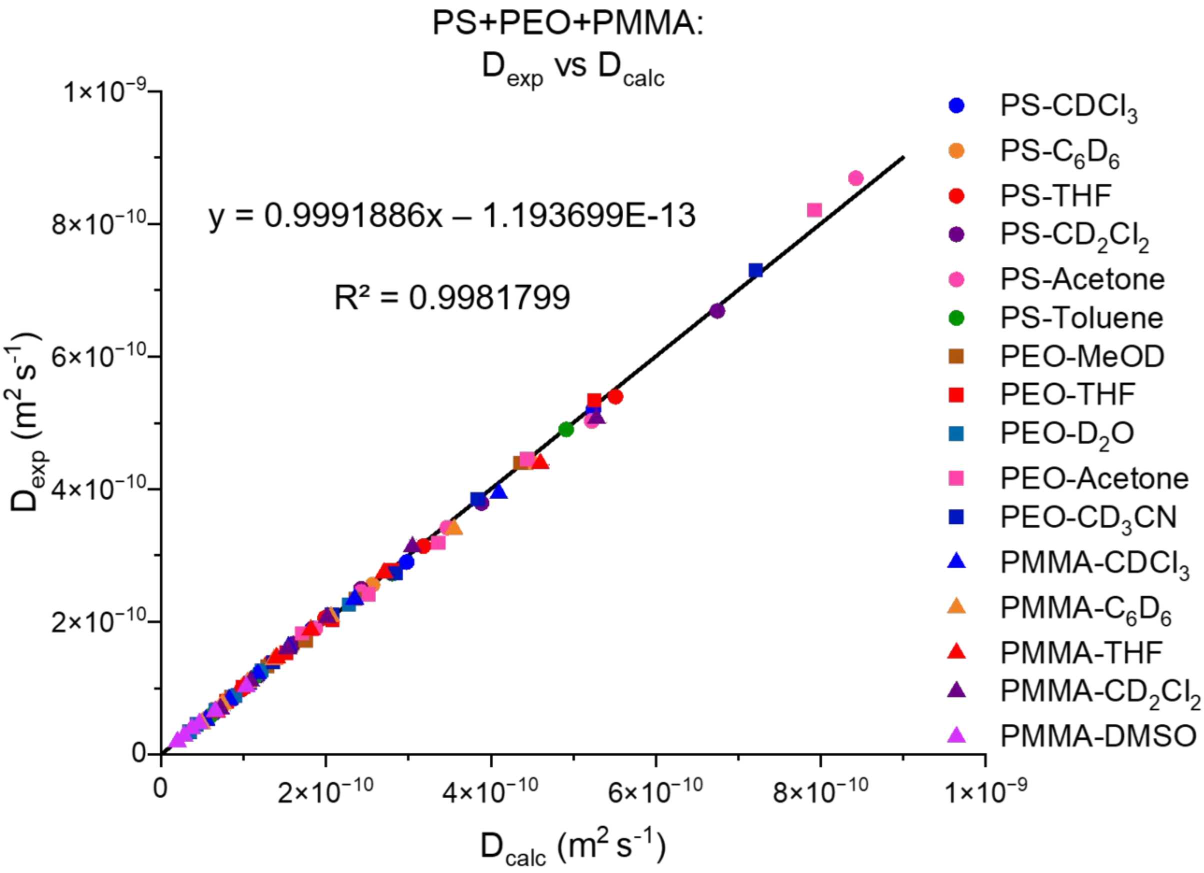

Figure S4: Experimental diffusion coefficients vs. calculated diffusion coefficients via equation (13) for PS, PEO and PMMA for linear axis.

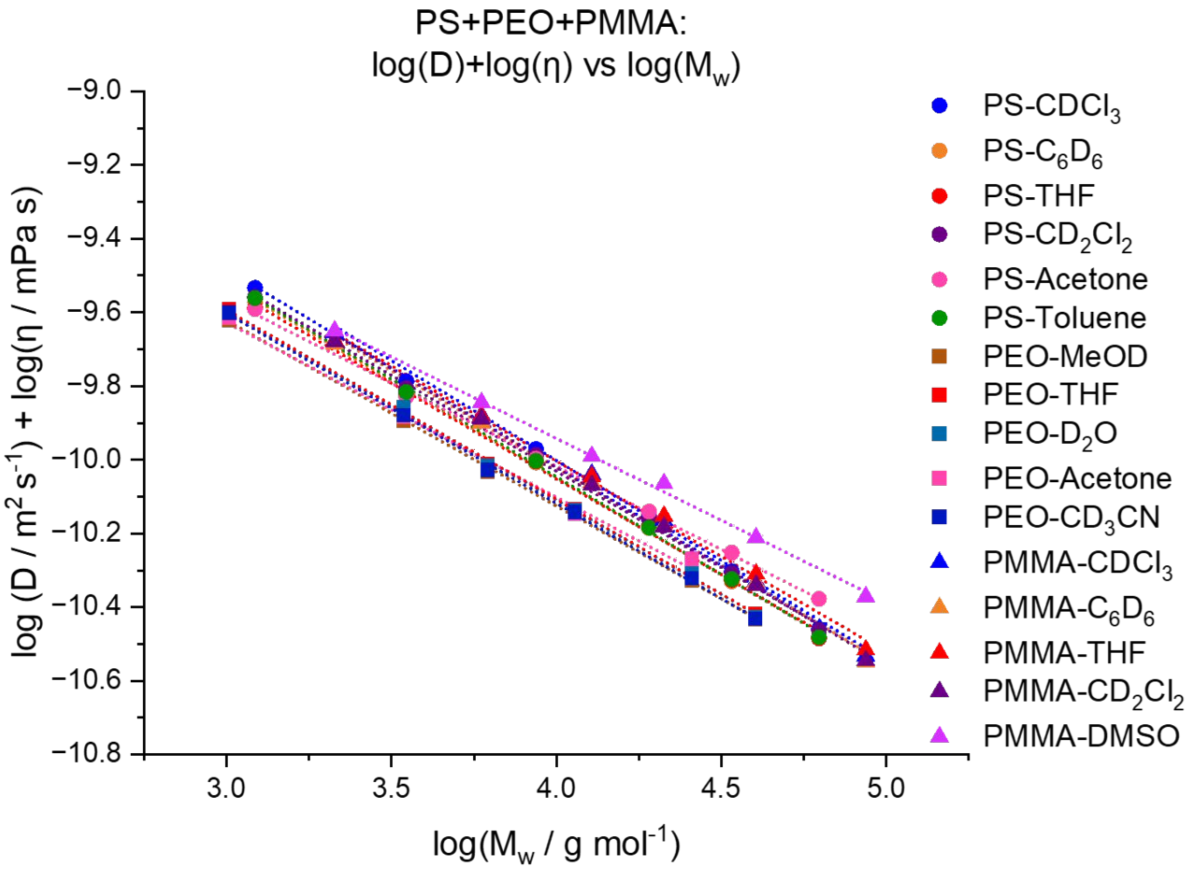

Figure S5: Dynamic viscosity corrected diffusion coefficients vs. molar masses for PS, PEO and PMMA in different solvents at T=298.15 K.

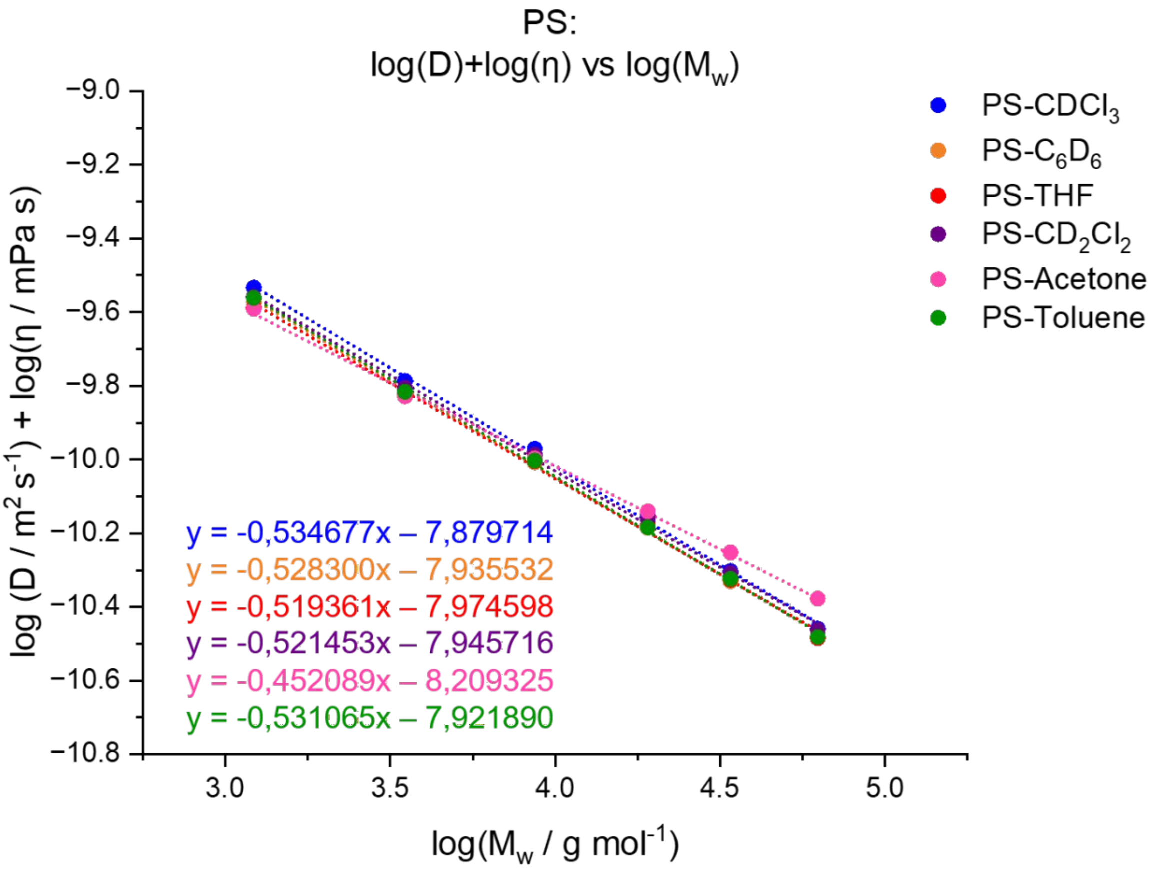

Figure S6. Dynamic viscosity corrected diffusion coefficients vs. molar masses for polystyrene in different solvents.

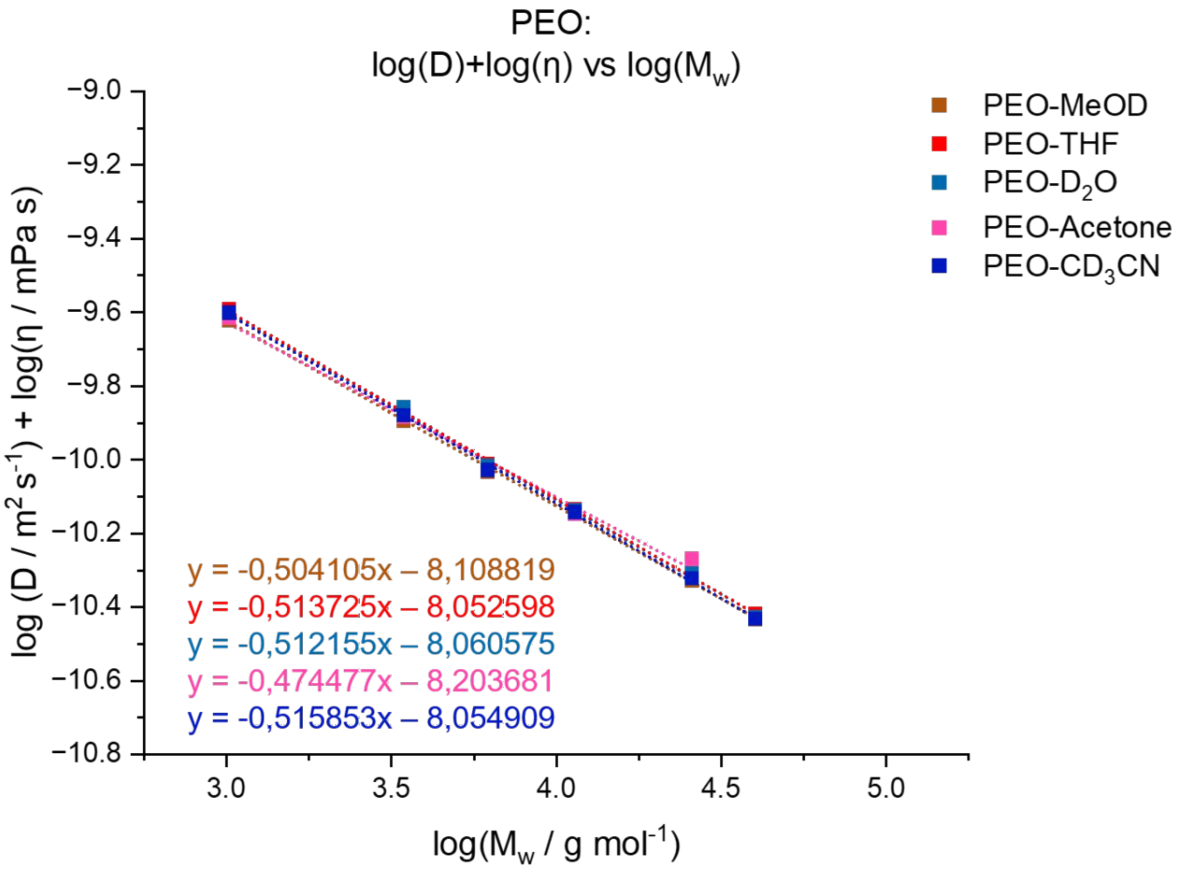

Figure S7. Dynamic viscosity corrected diffusion coefficients vs. molar masses for polyethylene oxide in different solvents.

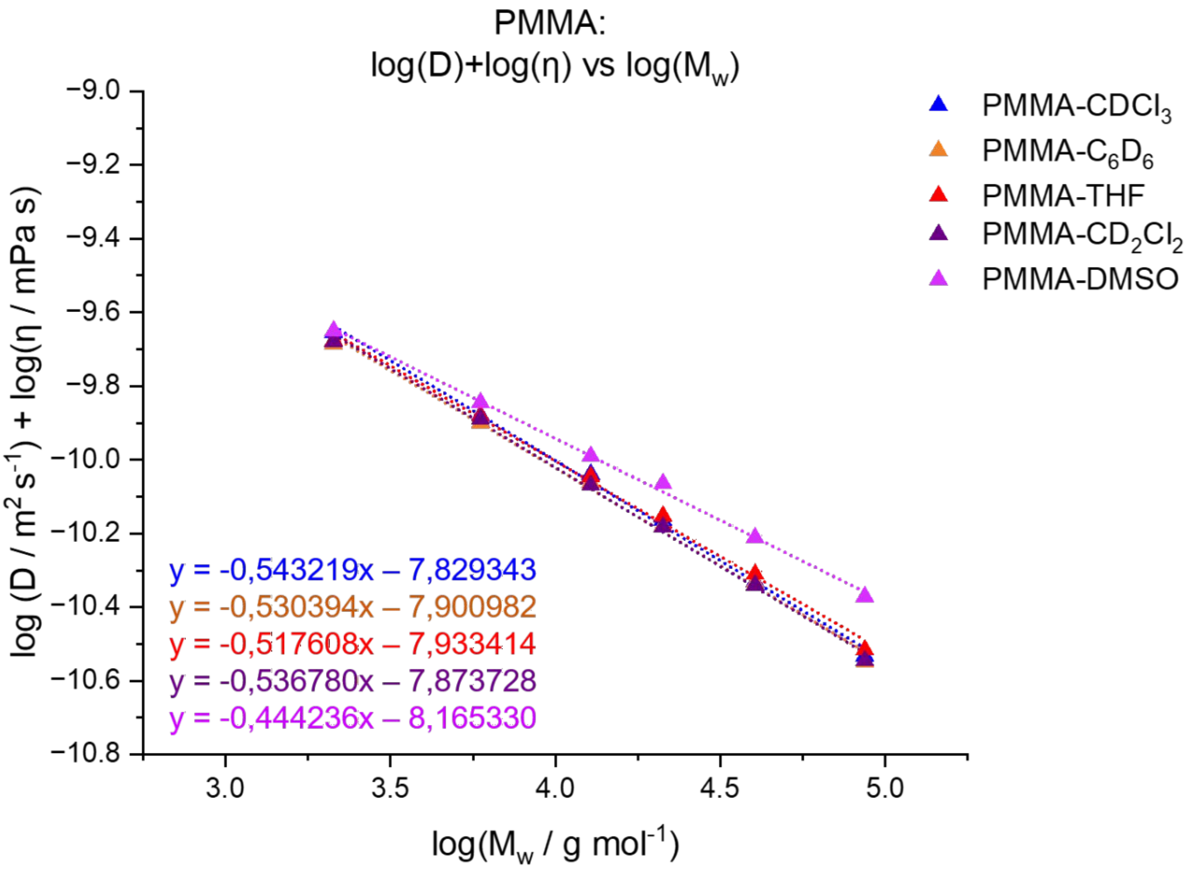

Figure S8. Dynamic viscosity corrected diffusion coefficients vs. molar masses for poly(methyl methacrylate) in different solvents.

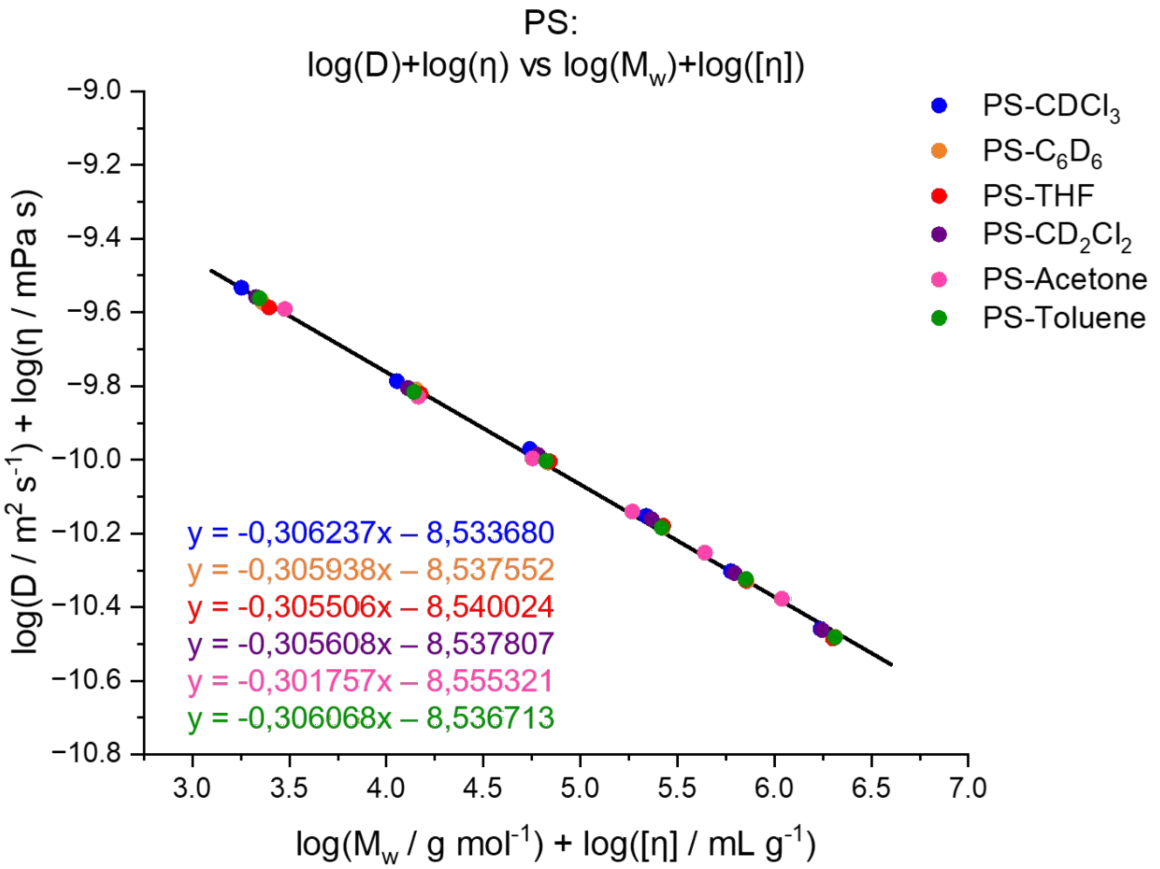

Figure S9. Dynamic viscosity corrected diffusion coefficients vs. intrinsic viscosity corrected molar masses for polystyrene in different solvents.

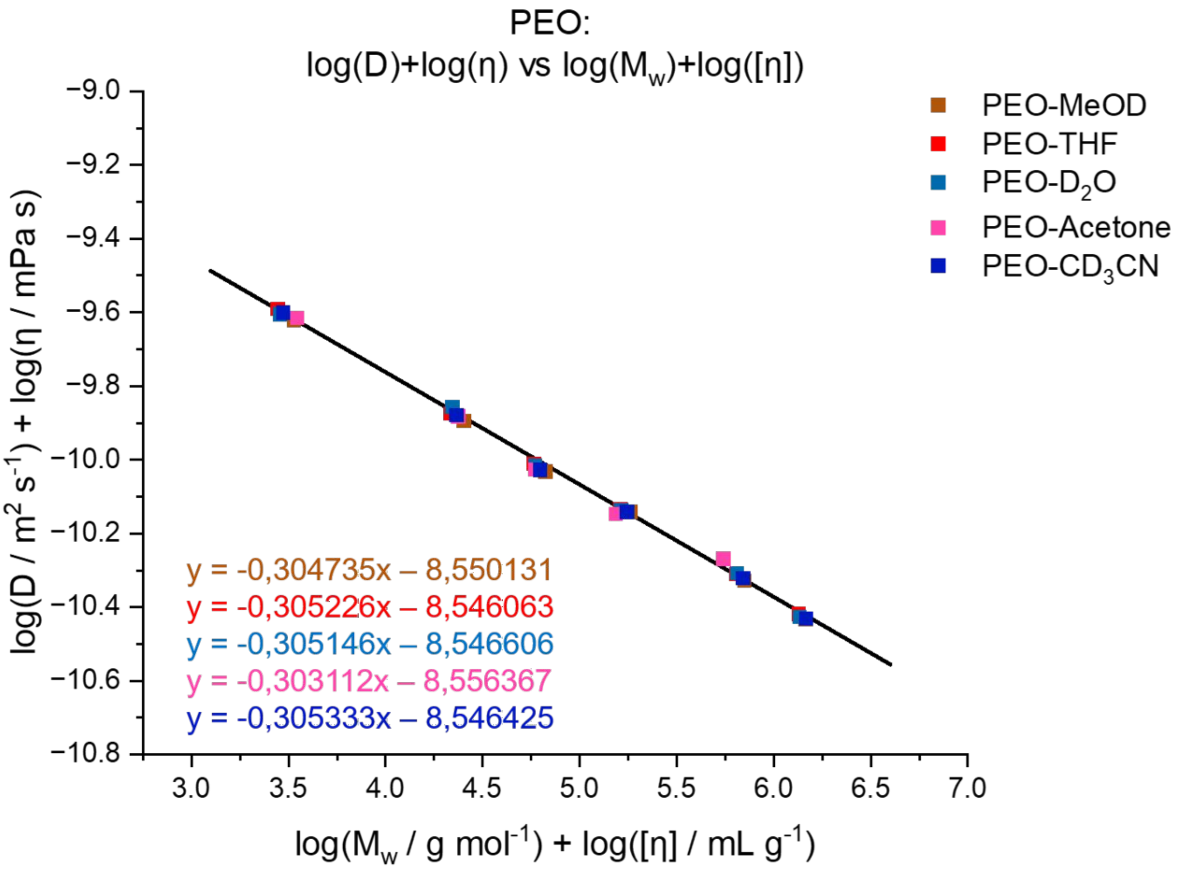

Figure S10. Dynamic viscosity corrected diffusion coefficients vs. intrinsic viscosity corrected molar masses for polyethylene oxide in different solvents.

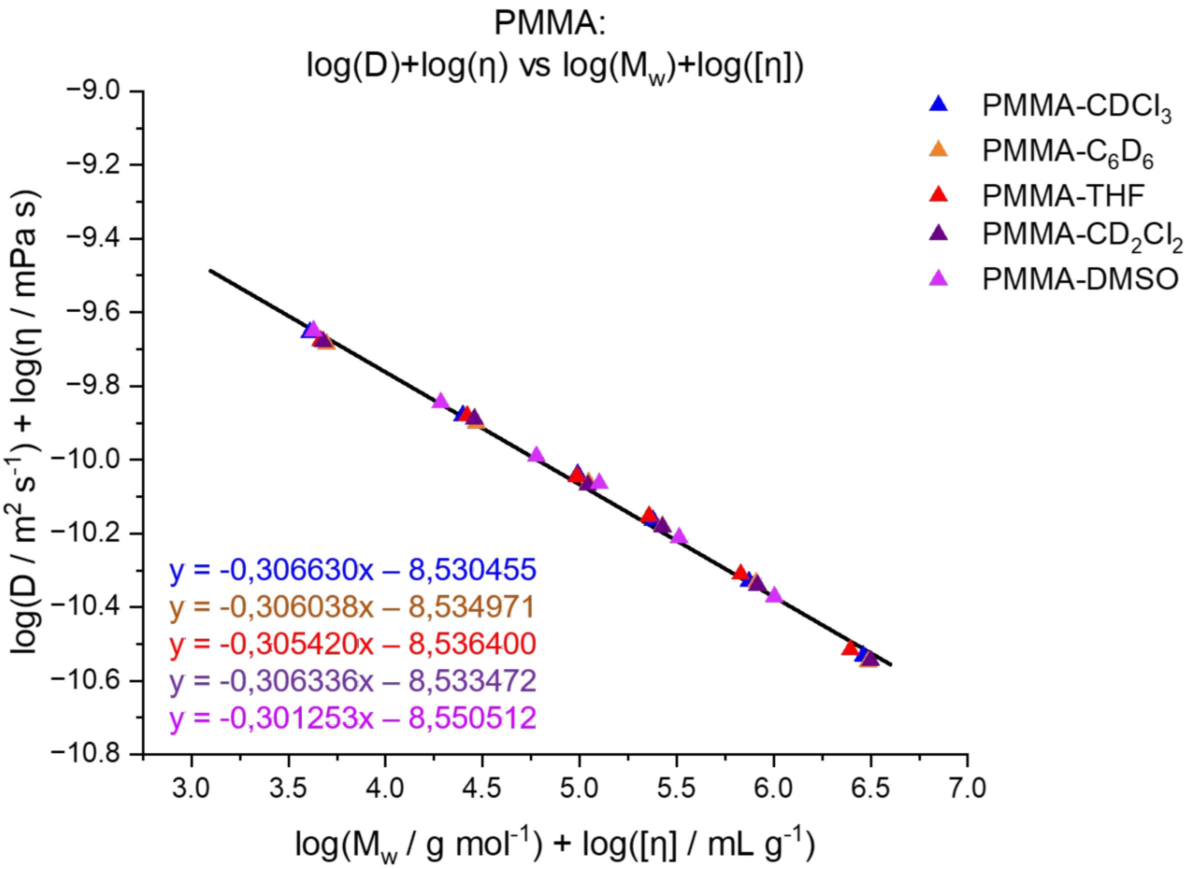

Figure S11. Dynamic viscosity corrected diffusion coefficients vs. intrinsic viscosity corrected molar masses for poly(methyl methacrylate) in different solvents.

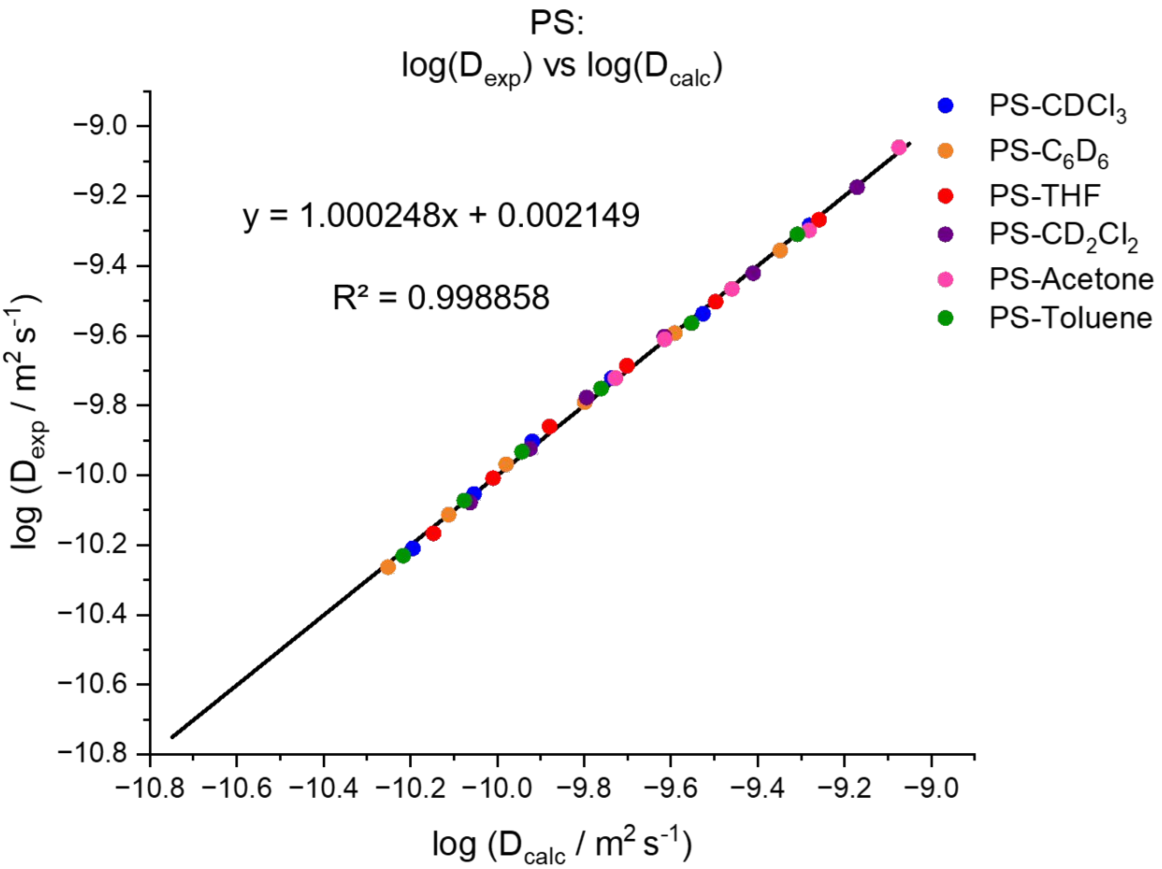

Figure S12: Experimental diffusion coefficients vs. calculated diffusion coefficients via equation (13) for PS.

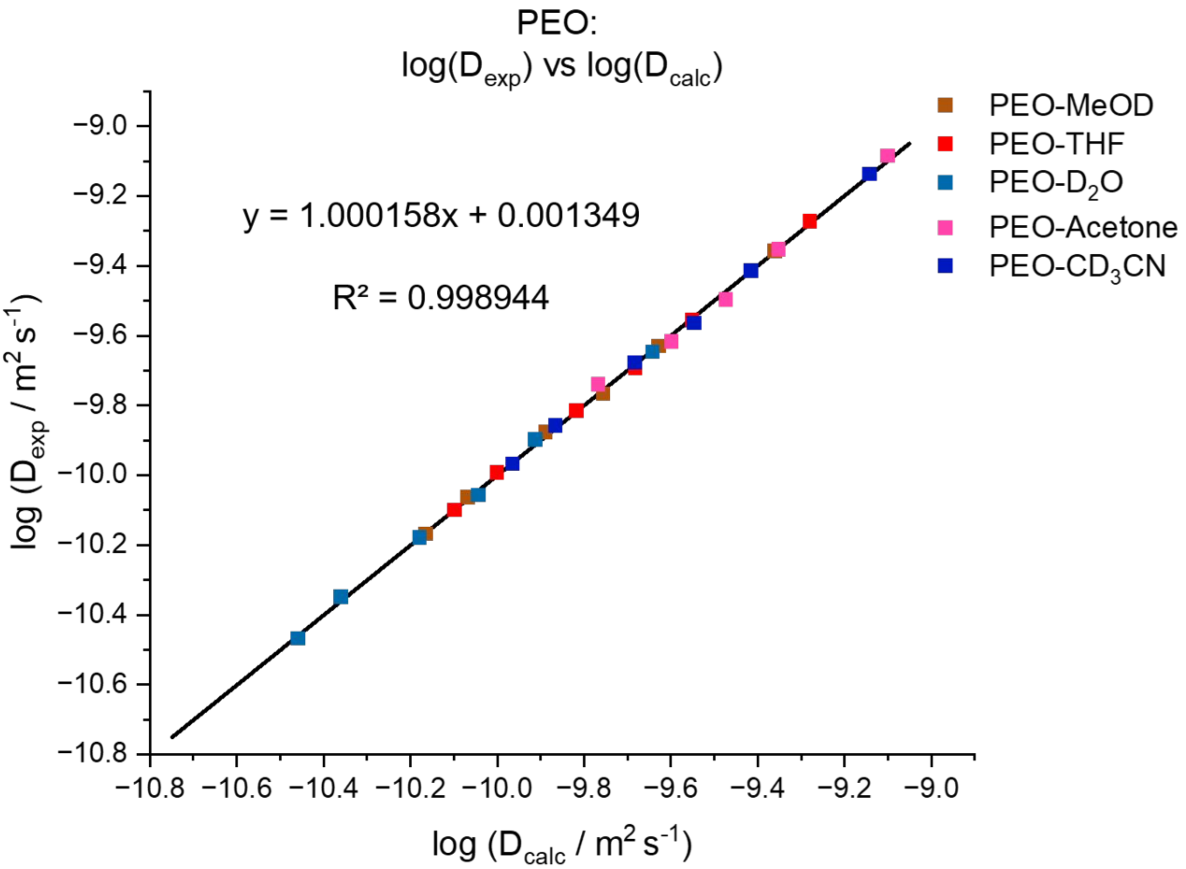

Figure S13: Experimental diffusion coefficients vs. calculated diffusion coefficients via equation (13) for PEO.

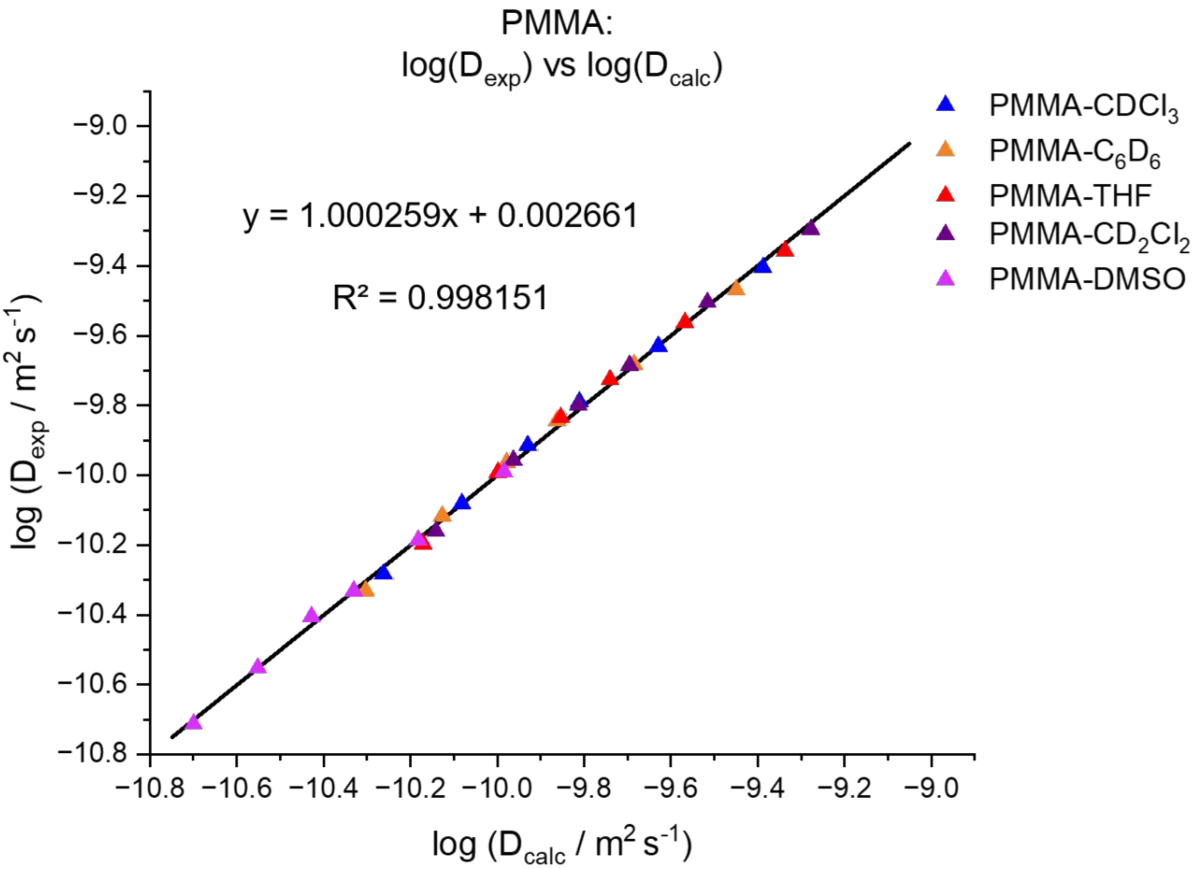

Figure S14: Experimental diffusion coefficients vs. calculated diffusion coefficients via equation (13) for PMMA.